

Natural Vegetation and Wildlife

- India is rich in biodiversity and is among the 12 megabiodiversity countries in the world.
- 4700 species of plant, and ranks 10th in the world in terms of diversity.
- 15000 species → ~~Flaming~~ Flowering plant.
- 90,000 species present in India.
- Natural Vegetation: It refers to plant community which has grown naturally without any human disturbance and is left undisturbed.
- ⇒ Wildlife: All the wild animals living in their natural environment.
- ⇒ Species may be:
 - 1) Indigenous Species.
 - Also called as native species.

- That naturally occur in a particular region or area, e.g. Bengal Tiger, Neam.

2) Exotic Species: Those species which are introduced from another area.

e.g. Cantara Camora.

3) Endemic Species: That are present in a particular geographical location.

e.g. Hangul. (J&K).

→ Flora: Species of plants are collectively called as flora.

→ Fauna: Species of animals are collectively c/a Fauna.

⇒ Types of Vegetation:

1) Tropical Evergreen forest.

2) Tropical deciduous forest.

3) Tropical thorn forest or scrubs.

4) ~~Mon~~ Mountain forest.

5) Mangrove forest.

1) Tropical Evergreen Forest

⇒ Location: Present in the areas of Western Ghats, Islands of Andaman and Nicobar, Upper parts of Assam and T.N Coast.

→ They flourish in area where rainfall is high. (more than 200 cm).

→ Characteristics: Height of trees is high (600 m or more).

→ They do not shed their leaves, thus forest remains green throughout the year.

→ It has all kinds of trees, shrubs, creepers, etc.

Thus multi-layered structure.

→ Commercially imp. Trees - Ebony, Mahogany,

Rubber, Rosewood, Cinchona.

→ Animals - Elephant, Monkey, Horned rhino (Assam W.B)

② Tropical Deciduous Forest

- Most Widespread forest.
- Also called as monsoon forest.
- Rainfall: 70-200 cm

- 2 Types -

1) Moist deciduous forest.

2) Dry deciduous Forest.

⇒ Moist Deciduous Forest	Dry Deciduous Forest
- <u>Rainfall</u> : 100-200 cm	- <u>Rainfall</u> - 70-100 cm
- <u>Location</u> - Eastern part of the country, N-Eastern states, along Himalayan foothill, Jarkhand, Odisha,	- <u>Location</u> - Rainer part of Peninsular Plateau, Bihar, UP, etc.

eastern slope of western
ghat.

- Plantation: Teak, sal, peepal,
Neem, etc.

- Plantation: Teak → dominant
specie.

Bamboo, sal, Mulberry,
kusum, khair →
Commercial plant.

→ Large part of these forest
are cleared for agriculture.

- Animals: Lion, tiger, deer, and elephant, variety
of birds, snakes, lizards, etc.

③ Thorn forest and scrubs

- Rainfall: Less than 70cm

- Vegetation consists of thorny trees and bushes,
to conserve water.

- Location: Present in N-Western part of India, in
the semi-arid area of Gujarat, Rajasthan, M.P.

Chettisgarh, V.P, Maryana, etc.

→ Plantation - Acacia, Palm tree, Euphorbia, and cactus.

- Scattered Vegetation, long deep roots to absorb water

- Stems are succulent (~~thick~~ thick and fleshy to store water).

- Leaves are thick to minimize water loss.

- Animals: rats, mice, rabbit, wolf, horses, camels, ~~the~~ lion, tiger, etc.

(4) ⇒ Mangrove Forest -

- Tidal forest, which are found in coastal areas influenced by tides.

- Location - They are found in coastal areas, ~~influen~~

→ Location - They are found on the deltas of Ganga, Mahanadi, Kaveri, Krishna, Godavari.

→ Plantation - Mangrove trees are dominant here.

They develop breathing roots outside the soil surface.

→ Ganga Brahmaputra Deltas - Sundari tree is present here. It provides durable timber.

- Palm, Coconut, Agar, etc.

- Animals Royal Bengal Tiger, Gharial, Snakes, etc.

(5) → Montane Forests:

• Vegetation found in mountainous areas.

→ As the altitude increases, ~~temp~~ temp. decreases and vegetation changes.

→ On the basis of altitude, vegetation is divided into following types:

1) The wet temperate forests:

- Altitude: 1000 - 2000 m.
- Present ~~in~~ on low hills.
- Evergreen, broad leaved trees: e.g., Oak tree and chestnut.

2) Temperate Forests

- ~~Altitude~~ Altitude: 1500 - 3000 m.
- Location- Southern slope of Himalayas, high altitudes of S. India, N-Eastern part of India.
- Vegetation: Pine, Deodar, Cedar, Spruce, Silverfir. (Coniferous Trees).

3) Alpine Forest

- Altitude: Above 3600 m.
 - It consists of Temperate vegetation (Pine, Silverfir) and grasslands.
 - As we move towards the snowline, shrubs, scrubs and grasses are present.
 - As we move above, lichens and mosses are ~~forest~~ present. (Tundra Vegetation).
 - Grasslands here ^{are} used by ~~Gujjar~~ Gujjars and bakormals for grazing.
- Animals: Kashmiri stag (Hangul), Spotted deer, Snow leopard, Wild sheep, Tibetan Antelope, Bear, Sheep, Goat, etc.

⇒ Wildlife

- India is rich in fauna species.

90,000 species of animals
2000 species of birds
2546 species of Fishes

→ consists of 13% of the World's species.
12% of the World's species.

- 5-8% of the World's amphibians, reptiles and Mammals belongs to India.

⇒ Important Wildlife:-

1) Elephant - Majestic mammal.

- Found in wet forests of Assam, Karnataka and Kerala.

2) One-horned rhinoceros:-

- Endemic to India.

- Found in marshy and swampy areas of Assam and W. Bengal.

3) Camel - Present in Thar Desert.

4) Lion - Present in Gir forests of Gujarat.

5) Tiger - Present in M.P., Sunderbans of W. Bengal, Himalayan region.

- India is the only country with natural population of tiger and lion.

6) Leopard - Belongs to cat family.

⇒ Fauna in Himalayas:-

• These animals have adapted themselves to survive in cold temperature.

- Ladakh: e.g. Yak, shaggy ~~horned~~ horned wild ox (weighs around 1 ton).

Tibetan Antelope, the Bharal (blue sheep), wild sheep, the Kiang, Ibex, Snow Leopard, Rare Red Panda.

→ India: Rich variety of birds as well.
e.g. Peacock, Pheasants, Ducks, Pigeons, etc.

→ Water: Crocodile, Gharial, Turtle and different species of fishes.

⇒ Importance of Flora and Fauna:

- 1) We get crops which are used as food source.
- 2) Medicinal plants are obtained from flora.
- 3) Milching animals, which produce milk is used for different purposes.
- 4) Animals are used for meat, eggs and wool.
- 5) Animals are used for transportation purpose.

- 6) Fish is used as a nutritious diet.
- 7) Insects help in pollination of crops, fruits and Flowers.

→ Threats to flora and Fauna:

- 1) Hunting and poaching by greedy hunters.
- 2) Pollution caused by industrial waste.
- 3) Acid Rain.
- 4) Introduction of Alien species.
(Which is introduced from outside).
- 5) Over-exploitation by humans.
- 6) Deforestation - large scale cutting of trees.
- 7) Land over change. (Forest land changed to agricultural land).